

Weiming Hu

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EDUCATION

Sichuan University	Chengdu, Sichuan, China
B. Eng in Civil Engineering	2016.9 - 2020.6
- First prize of Sichuan Province in National Mathematics Competition for College Students. 2017 Individual Second-class Scholarship. 2018 Individual First-class Scholarship.	
ShanghaiTech University	Shanghai, China
Master in Computer Science	2020.9 - 2023.6
Core Course: Computer Architecture II (A-), Computer Architecture III (A), Digital VLSI Design Project (A)	
Shanghai Jiao Tong University	Shanghai, China
Ph.D. in Computer Science	2023.9 - 2027.6 (Expected)

PUBLICATION

- Weiming Hu**, Haoyan Zhang, Cong Guo, Yu Feng, Renyang Guan, Zhendong Hua, Zihan Liu, Yue Guan, Minyi Guo, Jingwen Leng, 'MANT: Efficient Low-bit Group Quantization for LLMs via Mathematically Adaptive Numerical Type', to appear in IEEE International Symposium on High-Performance Computer Architecture (**HPCA 2025**)
- Cong Guo*, Jiaming Tang*, **Weiming Hu**, Jingwen Leng, Chen Zhang, Fan Yang, Yunxin Liu, Minyi Guo, Yuhao Zhu, 'OliVe: Accelerating Large Language Models via Hardware-friendly Outlier-Victim Pair Quantization', Proceedings of the 50th Annual International Symposium on Computer Architecture (**ISCA 2023**)
- Weiming Hu**, Yi Zhou, Ying Quan, Yuanfeng Wang, Xin Lou, 'Cache-locality Based Adaptive Warp Scheduling for Neural Network Acceleration on GPGPUs', IEEE 35th International System-on-Chip Conference (**SOCC 2022**)

INTERNSHIP

Glenfly Tech Co., Ltd. (Shanghai Zhaoxin Semiconductor Co., Ltd., GPU Department)	Shanghai, China
GPU Architecture R&D Intern, Core Pipeline Group	2021.8 - Now
- Assist the performance team to maintain and develop performance analysis tools, which used to visualize data and analyze the bottleneck by hardware counter of each GPC. - Research advanced GPGPU architecture, focus on warp scheduling, data management, L1 cache, etc.	

RESEARCH EXPERIENCE

GPGPU/GPU Architecture Design	Shanghai, China
- Review the topic about GPU architecture and read related paper. Design a statistical mechanism to quantify intra-warp locality and inter-warp locality of application. The proposed mechanism select warp scheduling policy according to locality information. - Implement it in GPGPU-Sim, evaluate the performance improvement under the proposed warp scheduling policy and write a manuscript.	

SKILL

Programming Languages: C/C++; Verilog, CUDA (novice); python, shell, Triton.

Simulator & Framework: GPGPU-Sim, Accel-sim, Sniper, GPUWattch, PyTorch.

Tool: Docker, Vim, GDB, Git, tmux.